AR-Based Intelligent Automobile Repair Learning & Evaluation System

RP26-IT-049

Project Proposal Report

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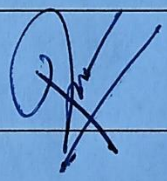
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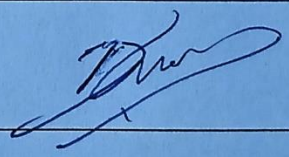
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Dr. Dharshana Kasturirathna	2026/03/16	

ABSTRACT

In traditional vocational training in automobile engineering, trainers often struggle with limited access to physical automobiles, the constant need for expert supervision by this trainer at every stage of a task and considerable safety risks during practical work. To address these gaps, we propose an AR-Based Intelligent Automobile Repair Learning & Evaluation System.

Spark AR positioning (the entire system is meaningful to identify components of assemblies' product, and the product can be assembled doing solutions such as Easy No-Mix and Nez) this module focuses on Practical Evaluation & Safety Feedback.

It automates student performance assessment by using an AI-based scoring algorithm and a Safety Rule Reasoning Engine as part of the proposed solution. Real-time data is entered into this system, such as accuracy of tasks done, time spent on a task, corrected mistakes from AR interactions etc.

The system uses Machine Learning (ML) to analyses performance and produce a quantitative performance score as well as a qualitative feedback report. Made safer with the safety engine, it includes real-time alerts on components that are at a high risk of misuse: such as warnings about high voltages when handling batteries.

Some of the technologies used here are AI scoring models and rule-based reasoning systems. Expected Outcome: A coherent and metric based personalized learning pathway including a globally recognized accreditation would significantly curtail human error hence bring down the costs associated with training while enhancing the employability quotient of students. By offering future automotive technicians a scalable, secure, and customized digital learning experience, this research advances quality education.

ACKNOWLEDGEMENT

I want to say thank you to everyone who helped me finish this research proposal.

First, I want to thank our supervisor, Dr. Dharshana Kasturirathna, for helping mean lot. He gave me good advice and encouraged me all the time. His knowledge of Information Technology was very important for this research proposal.

I also want to thank the Department of Information Technology at the Sri Lanka Institute of Information Technology for giving me what I needed to do this research. They gave me a place to work and the tools I needed.

I am also thankful to my group members (IT22284570,IT22231864,IT22279934) and friends for cheering me on and helping me stay on track. This research proposal is based on the work of other researchers, so I want to thank them too for doing such great work on Information Technology research. The research proposal would not have been possible without the research that these researchers did on Information Technology.

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LIST OF ABBREVIATIONS

- AI – Artificial Intelligence
- AR – Augmented Reality
- CNN – Convolutional Neural Network
- CV – Computer Vision
- DL – Deep Learning
- ID – Identification
- IEEE – Institute of Electrical and Electronics Engineers
- IT – Information Technology
- ML – Machine Learning
- SDG – Sustainable Development Goals
- SLIIT – Sri Lanka Institute of Information Technology
- UI – User Interface
- UX – User Experience
- WBS – Work Breakdown Structure

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Appendix	Description
Appendix A	“User Requirements Survey Questionnaire / Interview Guideline”
Appendix B	“Detailed System Architecture and Logic Flowcharts”
Appendix C	“Database Schema and Data Structures for Evaluation Logs”
Appendix D	“Safety Rule Matrix for Automotive Components”
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1. INTRODUCTION

1.1 Background and Context

Highly qualified technicians are needed due to the quick advancement of automotive technology, especially with regard to Electric Vehicles (EVs) and intricate engine management systems. However, there is a big gap in real-time practical learning because traditional vocational training frequently relies on manual supervision and static manuals. Due to restricted access to actual vehicle components and the high cost of damaged parts during training errors, students frequently struggle to obtain practical experience.

Furthermore, when it comes to auto repair, safety is a top priority. Fatal accidents can occur when handling high-voltage batteries or pressurized systems without prompt expert feedback. Automated systems that can impartially assess a student's practical performance while guaranteeing safety compliance in real-time are currently lacking. This study introduces an intelligent system based on augmented reality to address these issues. This module specifically focuses on offering a safety reasoning engine and a data-driven evaluation framework to ensure that trainees receive immediate feedback and risk alerts, improving workplace safety and learning effectiveness.

1.2 Literature Review

Augmented reality (AR) and artificial intelligence (AI) have recently been incorporated into vocational pedagogy to improve the acquisition of technical skills. While traditional e-learning platforms offer theoretical knowledge, they are unable to provide real-time feedback during in-person interactions with engine components, according to studies in automotive education. Current research lacks a standardized framework for automated practical evaluation and frequently concentrates on object recognition and instructional overlays.

Furthermore, the literature currently in publication still raises serious concerns about safety. A "Safety Reasoning Engine" that can anticipate and warn users about real-world hazards, like high-voltage risks in electric vehicles (EVs), is absent from many virtual training systems, despite the fact that they offer a risk-free environment. Previous research indicates that instructors' subjective and time-consuming manual evaluation results in inconsistent student grades. By combining rule-based safety monitoring with AI-driven scoring algorithms, this study fills these gaps and offers a more impartial, scalable, and secure evaluation model than conventional and existing digital solutions.

1.3 Research Gap

The absence of an automated, objective evaluation framework for real-world tasks is the main research gap in current automotive augmented reality applications. Although current systems are excellent at giving detailed instructions, they are unable to provide real-time safety reasoning during in-person interactions or quantitatively assess a student's performance. Most platforms provide static feedback, which does not simultaneously account for task accuracy, time efficiency, and safety compliance.

By using a Safety Rule Reasoning Engine and an AI-based Scoring Algorithm, my research fills in these gaps. By evaluating real-time data to produce customized performance reports and vital safety alerts, this system ensures a consistent and safe learning environment, going beyond basic guidance.

Identified Research Gap	How the Proposed System Addresses It
"Lack of objective scoring: Traditional methods rely on subjective instructor manual grading"	"Implements an **AI-driven scoring algorithm** based on accuracy, time, and sequence"
"Absence of real-time safety alerts: Current AR tools lack context-aware danger warnings"	"Features a **Safety Rule Reasoning Engine** that provides instant alerts for high-risk parts (e.g., Battery)"
"No personalized performance analytics: Students don't receive detailed mistake summaries"	"Generates a **Personalized Evaluation Report** with specific improvement suggestions"
"Fixed learning paths: Systems don't adapt to individual student errors"	"Uses **ML-based performance analysis** to identify and highlight recurring technical mistakes"

2. OBJECTIVE

2.1 Main Objective

To create an AI-powered Practical Evaluation and Safety Feedback system that ensures safety compliance during auto repair tasks through intelligent risk alerts and automated scoring while offering an objective, real-time assessment of student performance.

2.2 Specific Objectives

1. To create an AI-based scoring system that assesses student performance by examining task accuracy, action sequence, and time efficiency.

2. To put in place a Safety Rule Reasoning Engine that provides real-time preventive alerts and detects possible risks connected to engine components.

3. Creating a performance analytics module that monitors recurrent errors and offers a numerical analysis of a student's technical advantages and disadvantages.

4. To develop an automated feedback generation system that, following each practical session, generates an extensive evaluation report with tailored recommendations for enhancement.

5. To compare AI-generated results with manual evaluations carried out by automotive specialists to assess the scoring system's accuracy and usability.

3. METHODOLOGY

The Practical Evaluation & Safety Feedback module employs a methodical approach that combines a rule-based expert system with quantitative data analysis. Converting unprocessed interaction data from the AR environment into useful performance metrics and safety interventions is the main objective.

3.1 Gathering and Integrating Data

Initially, the system collects information from the modules that came before it, namely the Task Verification and Instruction units. Step correctness logs, time stamps for each action, and the Component ID (such as Battery) and Task ID (such as Inspection) are all included in the input.

3.2 AI-Powered Scoring System

The student's proficiency is assessed using a specially created scoring algorithm. Three primary areas are given weights:

• Accuracy (50%): Completing tasks in the right order and without omitting any steps.

• Time Efficiency (30%): A professional benchmark is compared to the user's completion time.

• Safety Adherence (20%): Any errors pertaining to safety will result in a mark deduction.

3.3 Reasoning Engine for Safety Rules

This engine uses a framework based on logic. The Reasoning Engine initiates a real-time alert if a student tries to interact with a high-risk component (such as a hot radiator or a high-voltage battery) without taking the necessary safety precautions.

3.4 Report on Creation and Verification

Lastly, the system creates a customized performance report based on the data analysis. To verify accuracy and dependability, this methodology will be validated by contrasting AI-generated scores with manual assessments from automotive instructors.

Phase	Description	Technology/Tools
Data Processing	"Processing logs of task completion and errors"	Python, JSON
Scoring Logic	"Calculating weighted scores based on performance"	Machine Learning (Regression Models)
Safety Logic	"Implementing rule-based safety triggers and alerts"	Rule-Based Reasoning Engine / Expert System
Feedback Delivery	"Generating visual dashboards and PDF reports"	Flask/Django API, AR Overlay
Data Storage	"Storing historical performance for progress tracking"	Firebase / MySQL

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Synopsis of the Suggested Resolution

The Automobile Learning System's last intelligence layer is the Practical Evaluation & Safety Feedback module. To create an impartial assessment, it combines information from task verification and component identification. Real-time interactions are processed by architecture, which transforms them into structured performance metrics and safety alerts.

4.2 Description of the System Architecture Diagram

The architecture should be separated into three primary layers in my system diagram:

1. Input Layer: Gets information from Component 3 (Task Verification), such as:

o Task Accuracy (right/wrong steps).

o the amount of time needed for each task.

o Particular Component ID (such as Battery).

o Error logs (skipped steps or improper tool use).

2. Processing Layer:

o AI Scoring Engine: Determines the final score using a weighted algorithm.

Safety Rule Reasoning Engine: A knowledge-based system that compares an action to a safety matrix (e.g., If Component = 'Battery' AND Action = 'Touch' WITHOUT 'Safety Gloves' -> Trigger Alert).

o ML Performance Analyzer: Examines past data to find trends in student mistakes.

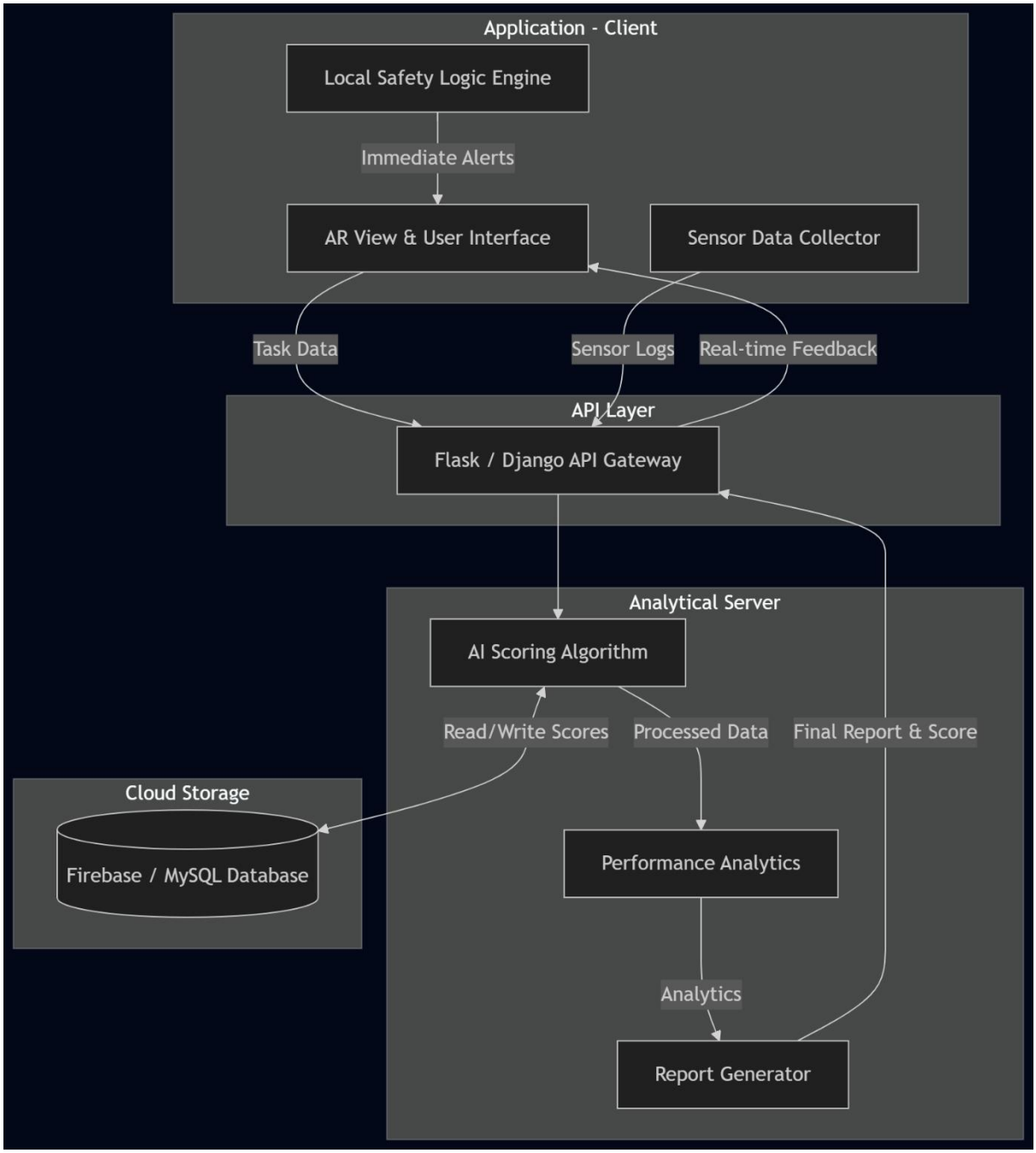
3. Output Layer:

o Real-time AR Safety Alerts: The user's screen displays visual alerts.

o Performance Dashboard: A graphic representation of the score.

o Detailed Evaluation Report: A summary with tailored suggestions for improvement that can be downloaded.

Data From	Processed By	Final Output
Step Correctness	AI Scoring System	Quantitative Grade (%)
Component ID & Action	Safety Reasoning Engine	Immediate Hazard Alert
Total Task Time	Efficiency Analyzer	Speed vs. Benchmark Rating
Mistake History	Feedback Generator	Personalized Study Plan


Application - Client
Local Safety Logic Engine
Immediate Alerts
AR View & User Interface
Sensor Data Collector
Task Data
Sensor Logs
Real-time Feedback
API Layer
Flask / Django API Gateway
Analytical Server
AI Scoring Algorithm
Read/Write Scores
Processed Data
Final Report & Score
Cloud Storage
Firebase / MySQL Database
Performance Analytics
Analytics
Report Generator

5. USER REQUIREMENTS

5.1 Identification of Stakeholders

The following are the main parties involved in the Practical Evaluation & Safety Feedback module:

• Students/Trainees: The final users who will carry out repairs and get safety alerts and real-time scoring.

• Teachers and assessors: Utilize the system to view automated evaluation reports and track students' progress.

• Vocational Institutes: Establishments aiming to lower safety risks and standardize their practical training.

5.2 Functional prerequisites (FR)

These outline the precise tasks that the system needs to complete:

1. FR-01: Real-time Performance Tracking: The system will monitor how accurately and in what order a student completes a task.

2. FR-02: Safety Alert Generation: If a student interacts in an unsafe way with a high-risk component (such as a high-voltage battery), the system will immediately issue visual or auditory warnings.

3. FR-03: Automated Scoring: Based on task completion, accuracy, and time spent, the system will compute a numerical score.

4. FR-04: Feedback Report Generation: The system will produce a visual dashboard or downloadable PDF that summarizes the students' errors and areas for development.

5. FR-05: Safety Rule Management: The system will enable the incorporation of safety procedures for various engine parts.

5.3 Non-Functional Requirements (NFR)

These characterize the system's quality attributes:

1. NFR-01: Real-time Response (Performance): To avoid possible mishaps, safety alerts must be sent out in milliseconds (low latency).

2. NFR-02: Accuracy: To guarantee fairness, the AI scoring algorithm needs to have a strong correlation with professional manual grading.

3. NFR-03: Usability: Beginners should be able to easily comprehend the feedback dashboard and AR alerts.

4. NFR-04: Reliability: Throughout a training session, the safety reasoning engine must always be accessible.

5. NFR-05: Data Security: Only authorized instructors should have access to student performance records and personal information.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The main target market includes schools that provide training, such as NAITA and VTA in Sri Lanka, private colleges that teach automobile engineering and big automotive service centers.

• Customer Persona: These are people who run schools and want to make grading safe and safety managers who want to prevent accidents during training sessions.

6.2 Value Proposition

• Objectivity: We replace manual grading with a fair AI-driven scoring system.

• Risk Mitigation: We help reduce costs and prevent accidents by alerting trainees to safety issues in time.

• Scalability: Our system allows one instructor to oversee students at once using a dashboard.

6.3 Revenue Model

• B2B Subscription (SaaS): Educational institutes pay an yearly fee based on the number of students.

• One-time Integration Fee: We charge for setting up custom safety rules for vehicle types, like electric vehicles.

• Premium Analytics: We offer data insights and reports for an extra fee.

6.4 Competitive Advantage

Our system is more than an instructional app. It. Measures performance and safety which is unique in the EdTech automotive sector.

6.5 Intellectual Property (IP) Considerations

• Patent: We can file a patent for the logic, behind our Safety Rule Reasoning Engine and Weighted AI Scoring Algorithm.

• Copyright: Our software code and the design of our evaluation dashboard will be protected by copyright laws.

Strategy	Details	Target Audience
Freemium	"Basic component identification & 2 tasks"	Individual students / Beginners.
Enterprise	"Full evaluation, safety alerts, & reports"	Training Centers & Universities.
Customized	"Specific safety rules for EV/Hybrid fleets"	High-end repair workshops.

7. BUDGET AND JUSTIFICATION

Category	Item	Cost (LKR)	Justification
Software	Unity, OpenCV, Python, TensorFlow/PyTorch, Git (all opensource)	0	"Open-source development tools used for AR, computer vision, and machine learning"
Hardware	1 Android smartphone for AR testing	35,000	"Required to test AR Core features and real-time camera processing"
Hardware	Tripod / phone holder for recording reference repair videos	3,000	"Ensures stable recording for dataset collection"
Cloud	Google Colab Pro (1 month) for ML model training/Cloud storage / Firebase for dataset storage	5,000	"Stores reference repair videos and system evaluation data"
Other	Mobile data for development and testing (12 months)	12,000	"Hard copies required for academic submission"
Contingency	Reserve for unexpected hardware issues	3,000	"10% contingency for hardware-dependent project"
TOTAL		**58,000**	

The project budget is kept relatively low because most development tools used, such as Python, OpenCV, TensorFlow/PyTorch, Unity, and AR frameworks, are free and open source. The main cost, totaling approximately LKR 58,000, is allocated for hardware devices required to test augmented reality features and real-time computer vision processing on physical mobile devices. Additional minor expenses include cloud services for machine learning experiments, mobile data for development activities, and printing costs for documentation.

8. WORK BREAKDOWN STRUCTURE (WBS)

The project is split into a few parts to make sure we have a good balance of work and can finish the Practical Evaluation & Safety Feedback module on time.

8.1 Project Phases and Task Breakdown

Phase 1: Research and Requirement Analysis happen in the two months

* We will read about how AIs used to score and keep people safe in car training.

* We need to figure out what the users need from this project both in terms of what it can do and how it should work.

* We must make safety rules for parts of the engine like the Battery and Radiator.

Phase 2: System Design and Logic Development happen in the next two months

* We will design the plan for the system.

* We will work on the logic for the Safety Rule Reasoning Engine.

* We will design an algorithm for the AI-Based System that gives more weight to some things.

* We will design the User Interface for the feedback dashboard, so it is easy to use.

Phase 3: Development and Implementation happen over the four months

* We will connect the data from Component 3 which's Task Verification.

* We will write the code for the Scoring Engine using Python and machine learning libraries.

* We will set safety alerts to happen in time in the augmented reality environment.

Phase 4: Testing and Validation happens in the two months

* We will test each part of the scoring logic and safety alerts to make sure they work.

* We will test the system with the other parts that our team members are working on.

* We will check the AI scores to make sure they are the same as what human experts would give.

Phase 5: Documentation and Final Submission happen in the two months

* We will finalize our research paper.

* We will get ready for the presentation and show off the product we made.

Member Name / ID	Main Responsibility	Key Tasks
IT21804960	**Practical Evaluation & Safety Feedback**	"Develop AI scoring algorithms, build Safety Reasoning Engine, generate evaluation reports, and implement real-time hazard alerts"
IT22279934	Vision-Based Identification	"Component detection using CNN/Computer Vision"
IT22231864	Task Recognition	"Predicting student tasks using ML classification"
IT22284570	AR Verification & Instruction	"AR overlay and real-time corrective guidance"

8.3 Timeline Alignment (Milestones)

• Milestone 1: We will complete the Literature Review and Proposal by Month 2.

The Literature Review and Proposal are really important, so we need to get them done on time.

• Milestone 2: The Safety Logic and Scoring Algorithm need to be finalized by Month 5.

This is a step and we must make sure the Safety Logic and Scoring Algorithm are correct.

• Milestone 3: All four modules must work seamlessly, and we plan to achieve this integration by Month 9.

The integration of all four modules is a milestone, and we will make sure all four modules are integrated properly.

• Milestone 4: Finally, we will submit the Final Research Project by Month 12.

The Final Research Project is the end goal. We will make sure it is complete and submitted on time as the Final Research Project.

9. GANTT CHART

Task	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Month →	Mar	Mar	Mar	Mar	Apr	Apr	Apr	Apr	May	May	May	May	Jun	Jun	Jun	Jun	Jul	Jul
1. Literature Review	■	■																
2. Requirements & Design			■	■														
3. Proximity Monitor					■	■												
4. Context Collector							■											
5. Decision Engine								■	■									
6. Emergency Handler										■								
7. Component Coordinator											■	■						
8. Integration Testing													■	■	■			
9. Evaluation & Metrics															■	■		
10. Report & Documentation																	■	■

March 2026 – July 2026 | 18 Weeks | ■ = Active week

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11.APPENDICES

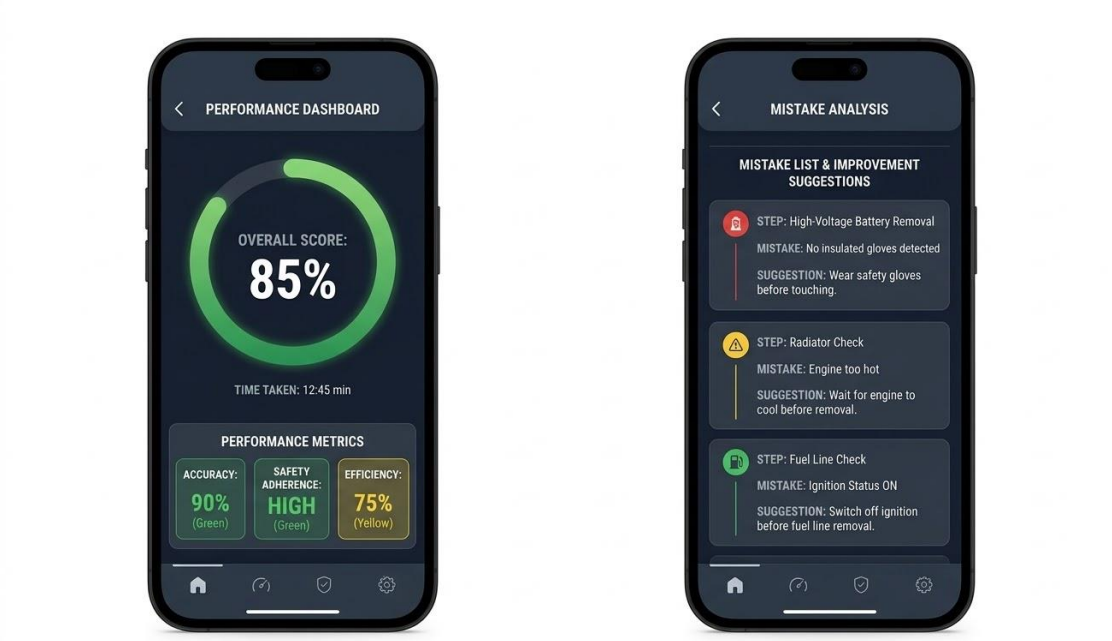


Figure D.1: Evaluation Dashboard UI Mockup

Appendix A: Initial UI Mockups: These are visual depictions of the AR interface that display the performance dashboard and safety alerts. The "If-Then-Else" conditions for high-risk automotive tasks (such as battery removal) are displayed in.

Appendix B: Detailed Safety Rule Set, a sample logic table.

Appendix C: Survey Instruments: Questionnaires created to find existing gaps in evaluation techniques for instructors and students studying automobiles. A summary of possible technical risks (such as AR latency) and their mitigation techniques can be found in.

Appendix D: Risk Analysis Table. Standard SLIIT ethical clearance forms for research involving student performance data are included in.

Appendix E: Ethical Approval Documents.

✓ Below is the similarity score obtained from Turnitin.

